

JSGP · SHINE HACKATHON

Morning Triage

The morning triage for One Care caseworkers:
who needs you first, and why, in plain language.

THE PROBLEM

Help that must be **asked for**

Alarm buttons and pendants only help while the fallen person can still press them. An unconscious senior presses nothing.

The fall becomes a **long lie**: hours helpless on the floor before anyone thinks to check.

THE APPROACH

Help that needs **no button**

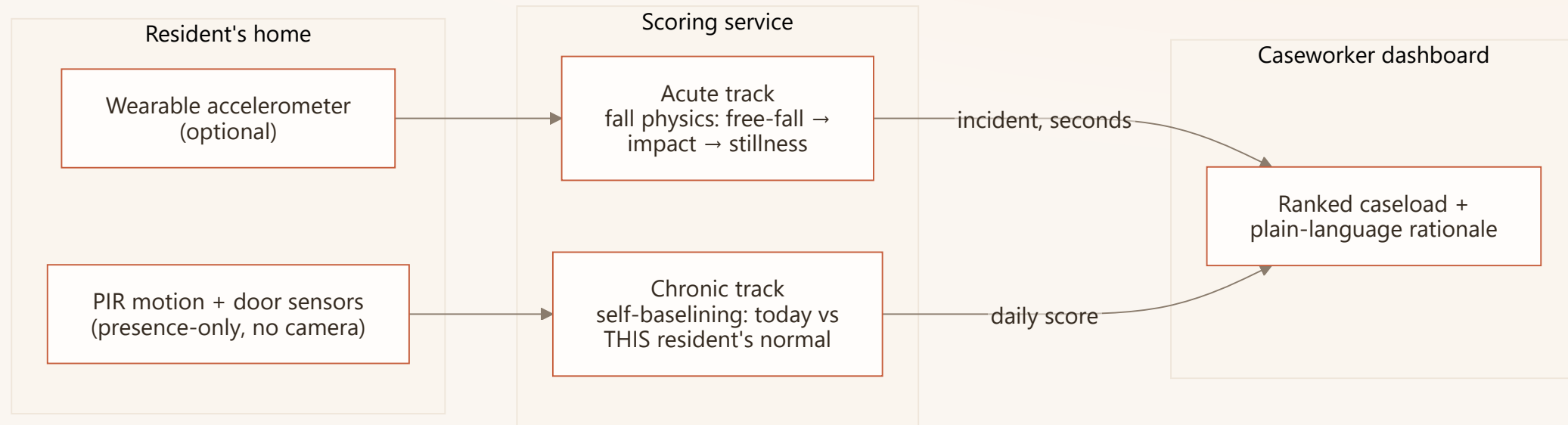
Motion and door sensors log every kitchen visit and every door event. When a morning breaks that pattern, or hours pass with no movement at all, the resident surfaces on the caseload. Nothing worn, nothing pressed, and no camera in the room.

One optional bangle adds a second layer:
falls caught in seconds instead of hours.

All of it is off-the-shelf hardware, and Singapore already deploys senior alert wearables at 26,800-resident scale.

HOW IT IS BUILT

The three layers



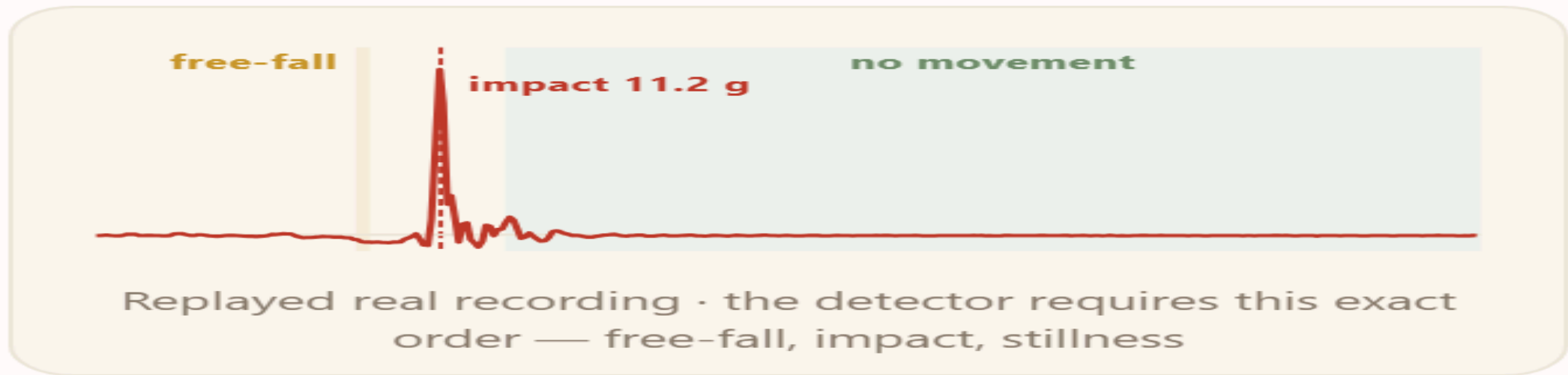
The home senses. The service scores. The caseworker decides.

The math is Python, so the service computes and the browser only displays. The API call carries the JSON between them.

HOW IT WORKS

A fall has a **signature**

WHAT THE SENSOR SAW



Straight from the product: the drill-down shows the signal behind every incident

The detector requires this **ordered sequence**: a free-fall dip, an impact spike within half a second, then stillness. A dropped phone cannot fake the order.

THE TWO DATASETS

Why these two recordings

4,505

REAL RECORDINGS · 1,798 FALLS · 38 PEOPLE · SISFALL

220 days

ONE SENIOR, LIVING ALONE · CASAS ARUBA

One teaches the detector what a fall is in physics.

The other teaches the system what one person's normal life is in rhythm.

Both run on the same cheap sensor class a worn bangle carries, the **ADXL345** accelerometer, so what the detector sees in the data is what it would see in a home.

WHERE THE NUMBER COMES FROM

How we got our **number**

- We wrote the detector first, the dip, spike, stillness rule.
- Then we ran it over all **4,505** recordings and **counted**: falls caught, and ordinary activities that false-fired.
- Then we swept **144** threshold combinations and kept the operating point that caught **96.2%** of the falls.

The number comes from counting an experiment. We measured a detector and did not train a model, so it carries to Singapore with no local data.

WHAT THE NUMBERS SAY

We ran it on **real recordings**

96.2%

OF 1,798 REAL FALLS DETECTED · SISFALL

35.2% → 18.8%

FALSE ALARMS: YOUNG ADULTS → ELDERLY MOVEMENT

220 days

ONE REAL HOME · CASAS ARUBA

The detector caught 96.2% of the 1,798 real falls. The false alarms split by body: 35.2% on young adults

hurling themselves into lab falls, but **18.8%** on the elderly participants, roughly half, because older movement is slower and less abrupt. On elderly falls the detector caught **84.0%**, though all 75 come from a single volunteer, so we hold that figure lightly. Nothing is trained here, because the thresholds came straight out of the data.

THE COUNTER-EXPERIMENT

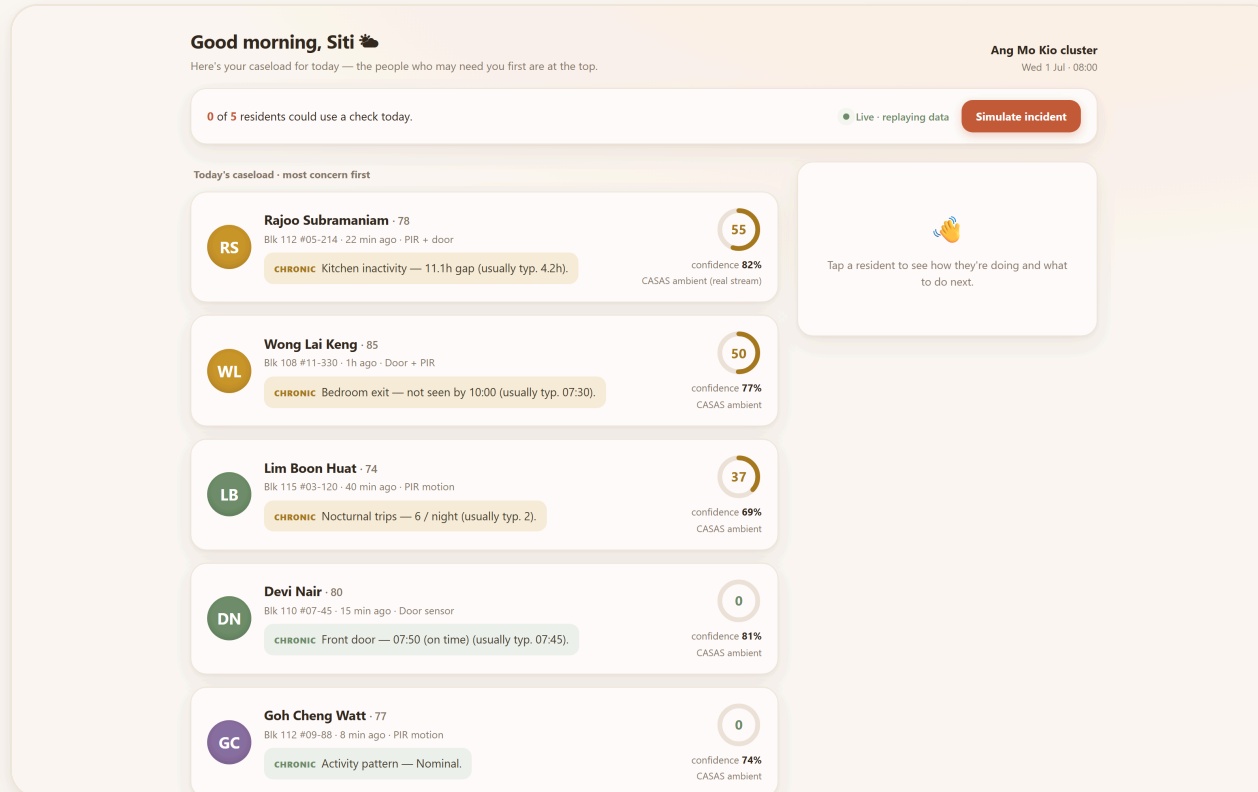
We trained a model **anyway**

- We fit a logistic regression on a curated table of **500** real SisFall recordings, 250 falls and 250 ordinary activities, four magnitude features per recording.
- On the held-out **125** recordings it reaches **80.0%** accuracy, 77.9% precision and 84.1% recall. Re-fits at 60/40, 70/30 and 80/20 move accuracy only between 78.5% and 80.0%.
- One fit takes **0.050 seconds**, so the `/training` page replays the real recorded gradient-descent run live.

The 16.2-point gap to the shipped detector's 96.2% is the evidence that the temporal order, free-fall then impact then stillness, is what magnitudes alone cannot see.

THE PRODUCT

One screen, calm by default



Every resident is ranked by need, and every row explains itself in one sentence.

EXPLAINABLE BY CONTRACT

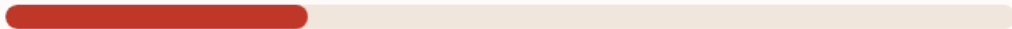
Every number **decomposes**

WHY SHE'S RANKED HERE

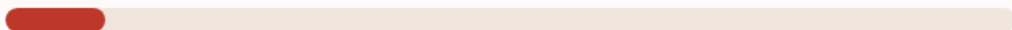
Peak impact **11.2 g**
walking < 1.4 g



Post-impact stillness **12 s**



Orientation **Horizontal**



Every score breaks into the sensor facts that produced it. Nothing is taken on faith

100
ACUTE RISK

100%
CONFIDENCE

Accelerometer
SOURCE

A stale sensor lowers confidence. It never inflates risk, so a dead sensor cannot fake a crisis

The system ranks and explains.

A human decides.

THE DEMO BEAT

A fall preempts everything

0 of 5 residents could use a check today.

● Live · replaying data

Simulate incident

Each press streams a different real recorded fall through the live detector, the same signal a worn bangle would send. A second button injects a dropped phone the detector ignores

FALL DETECTED

TA

Tan Ah Moi · 82
Blk 108 #04-210 · just now · Accelerometer

ACUTE Fall detected — 11.2 g impact then 12 s no movement.

confidence **100%**
Wearable bangle

Call now

Escalate

WHAT TO DO NEXT

CALL NOW. If no response, escalate to MOH 24/7 line + dispatch.

The fall pins to the top in seconds and states its evidence in plain words

The system suggests the next step. The caseworker makes the call

Two more live beats: `/watch` fires a named camera fall that answers back, and `/training` replays the trained-model run.

HONEST LIMITS

What we do **not** claim

- It shrinks time to detection. It does not promise to catch every event.
- The senior false-alarm rate is **measured** rather than guessed, because it came in at 18.8% on real elderly recordings, lower than young adults. What stays thin is elderly fall data, since our falls come from one volunteer, and a lab fall is not the same as a home fall.
- A missed fall (or a bangle left on the nightstand) degrades to the ambient track, where hours of stillness still surface that resident by morning.
- An irregular life earns wide baselines. The system says so through low confidence instead of hiding it.

THE NEXT STEP

What a pilot would **prove**

- **Escalation beyond the app.** The STILL DOWN follow-up and the I-am-responding acknowledgement are built and live. A pilot adds what sits past the phone: automated welfare calls, and the handoff to the MOH/AIC line when nobody acknowledges.
- **A morning brief per flagged resident.** Auto-drafted wording, grounded strictly in the deterministic features. The drafting layer never touches a score.
- **One One Care cluster to start.** Measured on two numbers: time to detection, and caseworker minutes saved per shift.

THE ASK

Pilot **with us**

We ask for one cluster and commodity sensors. The working system is already validated on **4,505 real recordings** and a **220-day real home**.

Built at SHINE Hackathon. The demo runs live on real recordings, a different one each press.

Why no machine learning

- **No Singapore training data exists.** A model trained on Colombian or American recordings would import exactly the prior we refuse to assume.
- **A fall is physics.** The free-fall, impact and stillness sequence transfers across bodies and homes without any training step.
- **A caseworker must be able to ask why.** Decomposable rules survive that question. A black box does not.
- Where a model helps later: smoothing the wording of the morning brief. It will never produce a score.

Where the **data** goes

What each dataset teaches is earlier in the deck. This is the provenance behind those numbers.

DATASET	SOURCE	PROVENANCE
SisFall	Universidad de Antioquia	4,505 recordings, 1,798 real falls, checksum-pinned
CASAS Aruba	Washington State University	220-day single-resident home log, checksum-pinned

Nothing is trained on either dataset. Both are provenance-pinned with checksums in a committed lock file, and every number reproduces from one script.

From a fall to the **dashboard**

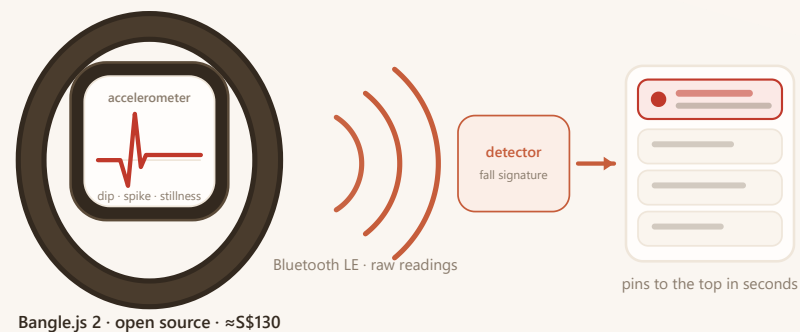
The path is the same in the demo and in deployment. Only the source of the signal changes.

1. The bangle streams accelerometer readings to the scoring service.
2. The detector watches every reading for the ordered fall signature: dip, spike, stillness.
3. A detection pushes an incident event down a live stream the dashboard is always listening to. The resident pins to the top in seconds.

Step 1 is the one link we simulate; there is no wristband in the room.

Each press of the Simulate button feeds a different real SisFall trace straight into step 2; a second button injects a dropped phone the detector ignores. Everything after (detection, push, re-rank) is the production path.

The hardware exists today



Live in the dashboard today: /hardware streams a recorded fall through the production path

DEVICE	REAL PRODUCT, BUYABLE NOW	PRICE
Wearable: raw accelerometer, BLE	Bangle.js 2 , Kionix KX022, open firmware	≈S\$130
Room presence · door events	Aqara P1 motion · contact, 5-yr battery	≈US\$20 · 15
Singapore precedent	GovTech WAAS (iWOW, 2025): 170 blocks · 26,800 seniors	govt tender

The camera layer, **named and opt-in**

- On `/watch`, MediaPipe pose runs entirely in the browser. Upright, then horizontal within **3 s**, then **3 s** of stillness fires the same incident path. A slower transition is a deliberate lie-down and never alarms.
- Every camera incident carries the label **Camera (pose)** and the heuristic's own 0.55 to 0.85 confidence band. The 96.2% figure belongs to the accelerometer track alone.
- Enrolled face identity is opt-in and on-device, matched only while the person is upright, so the caseload row and the Telegram ping are **named** and the alert says where to look: last motion: Bedroom. Unmatched people fire the generic alert.
- **The alert answers back.** No recovery 45 s after the alert sends STILL DOWN; a caregiver taps **I am responding** and the alert is stamped with who took it.
- No video leaves the browser; only the detection event and the matched resident id are sent.

BACKUP · FOR Q&A

What a flat **needs**

- Motion sensors in the rooms that carry routine: kitchen, bedroom, bathroom, living room.
- One contact sensor on the front door.
- One optional bangle for the acute track. The routine track works with nothing worn at all.
- An installer fills in a one-page sensor-to-area map. No cameras, no microphones, no rewiring.